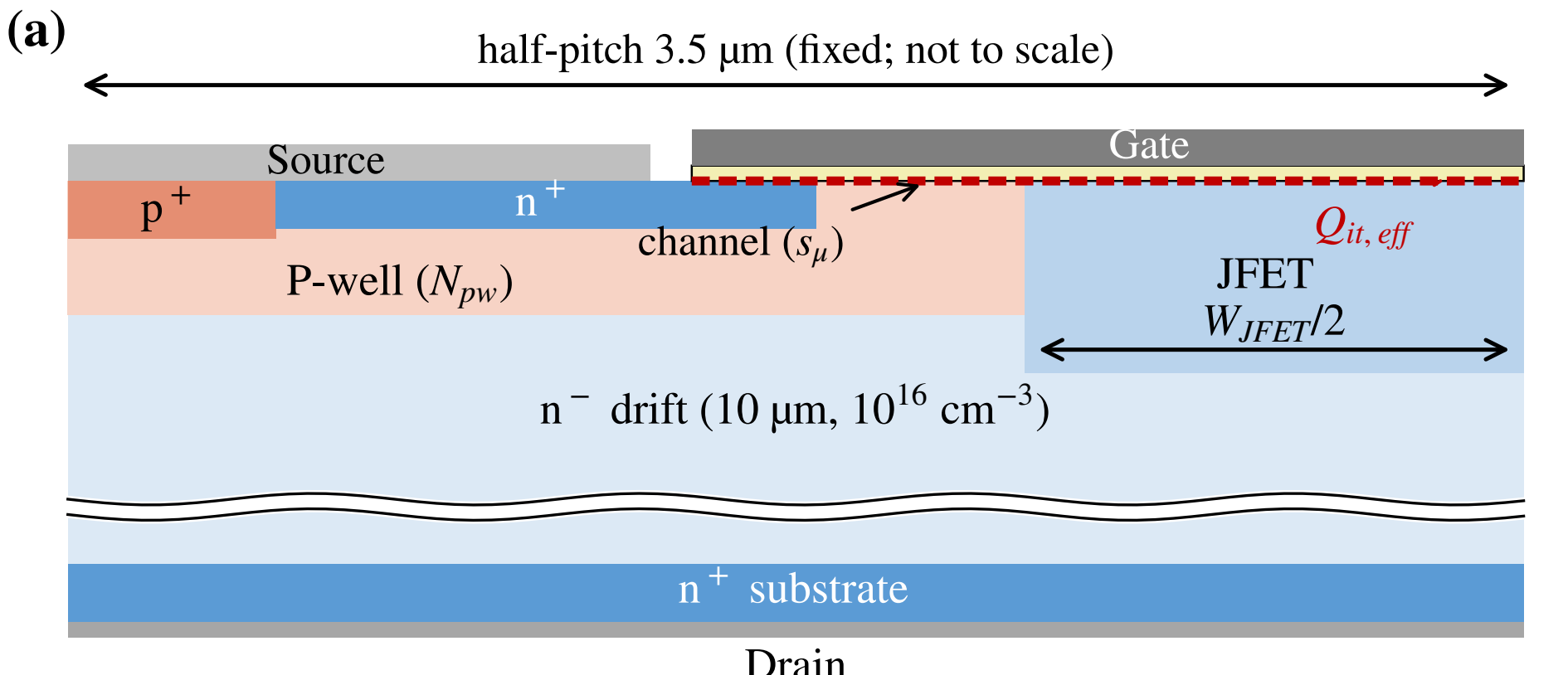
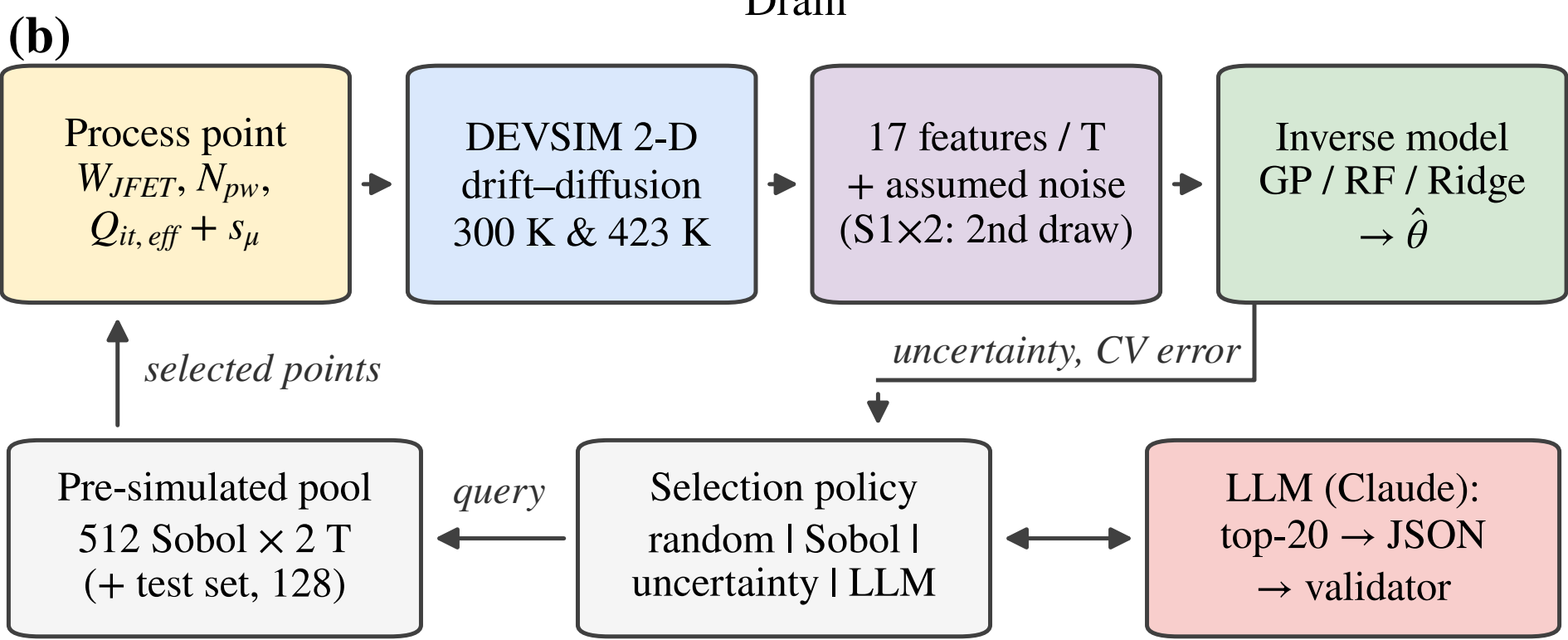

(a)
half-pitch 3.5 µm (fixed; not to scale)
Source
Gate
p+
n+
channel (sµ)
Qit, eff
P-well (Npw)
JFET
WJFET/2
n− drift (10 µm, 10^16 cm^−3)
n+ substrate
Drain


(b)
Process point
WJFET, Npw,
Qit, eff + sµ
DEVSIM 2-D
drift–diffusion
300 K & 423 K
17 features / T
+ assumed noise
(S1×2: 2nd draw)
Inverse model
GP / RF / Ridge
→ θ̂
selected points
uncertainty, CV error
Pre-simulated pool
512 Sobol × 2 T
(+ test set, 128)
query
Selection policy
random | Sobol |
uncertainty | LLM
LLM (Claude):
top-20 → JSON
→ validator